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Interplay of immersion, absorption, and flow in virtual reality: Based on experience economy theory and flow theory

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ABSTRACT

This paper investigates whether immersive experiences (escapism and esthetic experiences) precede and influence absorptive experiences (educational and entertainment experiences) in a VR tourism setting. In addition, we examine the mediating role of flow – characterized by enjoyment, concentration, and time distortion – in the relationship between immersion and absorption. An empirical investigation was conducted using a survey of 350 participants who took part in VR tours. Partial least squares structural equation model (PLS-SEM) was used to test the hypotheses. The results reveal that immersive experiences positively influence absorptive experiences when both experiences share the same level of participation, whether active or passive. Additionally, our study confirms that flow mediates the relationship between immersion and absorption, no matter the level of participation. This study is the first to explore a reverse causal relationship between absorption and immersion, challenging the existing literature. Additionally, it makes a valuable contribution by examining the mediating effect of flow within the framework of experiential economic theory in the context of VR tourism.

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Absorption; immersion; experience economy theory; flow theory; VR tourism

1. Introduction

The Experience Economy theory, first introduced by Pine and Gilmore (1999), has gained significant attention as it provides a framework to understand how businesses can create value through memorable experiences. According to this theory, consumers do not merely purchase goods or services; they seek engaging experiences that offer a deeper connection. Pine and Gilmore categorized experiences into four dimensions – entertainment, education, escapism, and aesthetics – based on the level of consumer participation and involvement. This theory highlights the importance of creating immersive and absorptive experiences that can foster stronger emotional connections and drive consumer satisfaction and loyalty. Recent studies have continued to explore the relevance of the experience economy in various fields, especially within virtual environments, such as fashion and tourism. For example, VR fashion marketing (Alexander & Rose Thompson, 2024; Herrera et al., 2020; Kim et al. 2022) and VR tourism (Han, Bergs, et al., 2022; Han, Boerwinkel, et al., 2022; Jung et al., 2021; Jung et al., 2016) have become

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a growing area where the four dimensions of experience play a vital role in shaping user satisfaction and behavioral intentions. In the literature, the experience economy theory has been applied in various contexts, including luxury fashion stores, cruises, bed and breakfast, music festivals, temple stays, and museums (Alexander & Rose Thompson, 2024; Mehmetoglu & Engen, 2011; Oh et al., 2007; Song et al., 2015). These studies focused on the four dimensions of consumer experiences and investigated how each dimension affects consumer outcomes such as satisfaction and behavioral intentions across different settings (Jung et al., 2016; S. Lee et al., 2020; Lin & Jun, 2022; Song et al., 2015; Sotiriadis, 2017). The findings indicate that the impact of each experience dimension on these outcomes may vary depending on the specific offerings in different settings (Oh et al., 2007).

Despite numerous studies exploring the relationships between the experience economy and satisfaction, few have examined the interactions between the four dimensions of consumer experiences (H. Lee et al., 2020; Park et al., 2010). Park et al. (2010) demonstrated that educational, entertainment, and esthetic experiences significantly influence escapism in the context of film festivals. In contrast, H. Lee et al. (2020) focused on the concepts of immersion and absorption, finding that educational and entertainment experiences (absorption) significantly influence esthetic and escapism experiences (immersion) in a VR museum setting. These contrasting findings suggest that research on the relationships between the four dimensions of customer experiences is still in its early stages, warranting further investigation into the 4Es with alternative hypotheses. In this study, we explore the relationships between the 4Es, focusing on immersion and absorption similarly to H. Lee et al. (2020). However, we propose an alternative hypothesis: educational and entertainment experiences (immersion) may precede esthetic and escapism experiences (absorption). Specifically, in virtual reality (VR) environments, consumers often become immersed in content that captures their attention through sensory engagement and detachment from reality before they absorb information from the experience, as VR's key characteristics – full immersion and presence – allow consumers to perceive virtual experiences as authentic (Guttentag, 2010). By investigating the relationship between immersion and absorption in the context of VR tourism, we aim to uncover the underlying mechanism through which virtual reality experiences enhance consumer satisfaction and behavioral intentions.

Additionally, we propose that flow may play a mediating role between these immersive and absorptive experiences. Flow is considered a typical aspect of consumer experiences in virtual environments (Hoffman & Novak, 1996). Csikszentmihalyi (1997) theorized that flow consists of both cognitive and affective components: cognitive components serve as antecedents to flow, while affective components influence attitudes and subsequent behavioral intentions (Hoffman & Novak, 1996). Following this cognitive-affective mechanism, several studies have examined the mediating role of flow in tourism, particularly between travel motivation and tourist behavioral intention (J. Wang et al., 2015), and between involvement and satisfaction (Y. J. Lee et al., 2019). Q. Zhang et al. (2021) explored the mediating role of flow in the relationships between the four dimensions of experience economy and tourist behavioral intentions. However, to our knowledge, no study has yet explored the mediating role of flow between immersive experiences and absorptive experiences of experience economy.

Specifically, this study addresses the following questions:

(1) How do the four dimensions of consumer experiences in virtual environments influence consumer satisfaction and behavioral intentions?
(2) What role does flow play in the relationships between immersive experiences and absorptive experiences?

In the absence of prior research combining experience economy theory and flow theory, this study sheds light on how the four dimensions of consumer experiences interrelate within the virtual reality environment and how flow influences these relationships. Although this study focuses on VR tourism as a context, VR has also been effectively utilized as a marketing communication tool in the fashion and luxury industries. Therefore, our research questions provide an important direction and insights for the marketing field, where consumer experience is deemed essential. Additionally, we expect that the findings of this study will provide managerial guidance on how to successfully implement VR as a marketing communication tool to enhance consumer experiences and behaviors in experience-intensive industries, including beauty, fashion, and tourism.

The remainder of this paper is organized as follows. Section 2 presents literature review and hypothesis development. Sections 3 and 4 detail the methodology and results, respectively. Finally, Section 5 discusses the findings, their implications for marketing theory and practice, and future research directions.

2. Literature review and hypotheses

2.1. Absorption and immersion

In Pine and Gilmore's (1999) experience economy theory, the four dimensions of consumer experiences are distinguished by two key factors: the level of participation (passive vs. active) and consumer involvement (absorption vs. immersion). First, regarding participation, the entertainment and esthetic dimensions involve passive participation, where the consumer does not influence the experiential outcome. In contrast, the educational and escapism dimensions require active participation, with consumers playing a critical role in shaping the experience (Pine & Gilmore, 1999). Second, the 4Es are classified into two overarching concepts: absorption and immersion. Absorption refers to occupying an individual's attention by mentally engaging them in the experience, whereas immersion involves becoming physically or virtually part of the experience itself (Pine & Gilmore, 1999). Among the 4Es, educational and entertainment experiences are considered absorptive, focusing on knowledge acquisition and enjoyment, while esthetic and escapism experiences are considered immersive, involving sensory engagement and fantasy.

In the tourism literature, the 4Es are further defined as follows: Educational experiences (active absorption) involve "learning from events unfolding at the destination through interaction and active mental engagement" (Oh et al., 2007). Entertainment experiences (passive absorption), typically characterized by passive engagement, such as attending concerts or watching horse races, are often assessed in terms of "fun" or "entertainment" (Quadri-Felitti & Fiore, 2012). Esthetic experiences (passive immersion)

are defined by "the beauty expressed through elements like color, photographs, font style, and layout" (H. Lee et al., 2015), which directly impact emotions and pleasure (Chatterjee, 2014). Finally, escapism (active immersion) refers to "escaping from a routine environment, suspending the norms and values that govern daily life, or viewing life and society from a different perspective" (Oh et al., 2007). The desire to "escape reality" is a key motivating factor for both the use of virtual worlds (Verhagen et al., 2012) and travel (Kozak, 2002).

In their research on the relationship between absorption and immersion, H. Lee et al. (2020) argue that absorptive experiences generally precede immersive ones, suggesting that individuals are more likely to experience immersion when they possess prior knowledge or interest. Their study found that absorption significantly influences immersion, particularly in VR experiences related to industrial cultural heritage. However, H. Lee et al. (2020) also acknowledge in their paper that absorption and immersion often occur simultaneously, especially in VR environments, allowing for alternative perspectives on the precedence of absorption and immersion. Agarwal and Karahanna (2000) proposed that immersive factors, such as temporal dissociation, focused immersion, and curiosity, precede cognitive absorption. Similarly, Han, Boerwinkel, et al. (2022) found that in VR-based dining experiences, initial engagement is primarily driven by immersive elements, which subsequently lead to deeper absorption.

In this paper, we explore whether immersive experiences (escapism and esthetic experiences) precede and influence absorptive experiences (educational and entertainment experiences). We develop our hypotheses based on the insights from previous research.

Several studies have examined the impact of escapism on absorptive experiences (educational and entertainment experiences). Kim and Song (2022) found that highly immersive qualities of AR guide applications stimulate novelty and curiosity – key factors in capturing consumer interests and enhancing educational outcomes – thereby positively influencing both educational and entertainment experiences. Garwood (2020) and S. Y. Huang et al. (2020) showed that escape room scenarios, such as the Mathematics Digital Escape Room (DER), effectively engage students and enhance learning through active involvement. Similarly, Wiesemüller et al. (2023) found that immersive VR escape rooms motivate learners and increase their perceptual sensitivity. In addition, Han, Bergs, et al. (2022) underlined that VR experiences provide an avenue for escapism, significantly amplifying the entertainment value of experiences by fostering deeper immersion. Irimiás et al. (2021) suggested that the more one immerses themselves in a fantasy world, the greater the potential for enhanced entertainment value.

Based on these insights, we propose the following hypotheses:

H_1: Escapism experiences influence educational experiences.

H_2: Escapism experiences influence entertainment experiences.

Previous studies have also explored the relationship between esthetic experiences and absorptive experiences (educational and entertainment experiences). Han, Boerwinkel, et al. (2022) suggested that esthetic encounters capture users' attention, enhancing their receptiveness to learning and engagement. Parrish (2009) found that integrating esthetics

into learning materials positively influences students' emotional engagement and cognitive processing, leading to improved educational effectiveness. Additionally, Y. Wang et al. (2021) argued that the visual aspects of esthetics provide instant fun and enjoyment. Esthetics also have positive effects, such as fostering comfort and pleasure while relieving stress (Sweeney & Soutar, 2001). Q. Zhang and Xu (2020) demonstrated that the beauty of destinations enhances visitor experiences, highlighting the importance of esthetic encounters in creating memorable and enjoyable moments. Nasiri et al. (2024) further emphasized that the quality of esthetic elements significantly influences tourists' perceptions and enjoyment.

Based on these insights, we propose the following hypotheses:

H_3: Esthetic experiences influence educational experiences.

H_4: Esthetic experiences influence entertainment experiences.

2.2. Flow and its mediating role

Flow, a concept introduced by Csikszentmihalyi (1975), refers to a "holistic sensation that people feel when they are fully immersed in an activity". It has been applied across various fields, including education, sports, and digital media, to enhance learning, performance, and user satisfaction (Nakamura & Csikszentmihalyi, 2009). In the experience economy, creating conditions that foster flow can significantly boost consumer engagement and improve overall experiences.

Despite its widespread recognition, the measurement of flow remains a subject of debate (Choi et al., 2007; Y. J. Lee et al., 2019). Some researchers have treated flow as a unidimensional construct (Chang, 2013; H. C. Huang et al., 2017), while others have considered it a multi-dimensional construct (Ding et al., 2010; Koufaris, 2002). Additional studies view flow as a second-order construct comprised of several first-order constructs. For example, Chou and Ting (2003) conceptualize flow as a higher-order construct consisting of concentration, time distortion, and playfulness, while Sanchez-Franco (2006) describes it as consisting of enjoyment and concentration. In this study, we adopt the view of flow as a second-order construct composed of three first-order dimensions – concentration, enjoyment, and time distortion – to measure the flow variable.

To understand how immersive experiences (escapism and esthetic experiences) influence absorptive experiences (educational and entertainment experiences) in VR tourism, we explore the role of flow as a mediating factor. Previous research have found that both escapism and esthetic experiences can induce a state of flow and significantly affect it in tourism contexts (Lin & Jun, 2022; Zhang et al., 2021). Zhang et al. (2021) suggested that the artistic attributes of theatrical performance provide a variety of stimulates that can evoke escapist and esthetic experiences in the cognitive level. When individuals are attracted to and immersed in a performance, they achieve a flow experience in the affective level. Wang et al. (2021) demonstrated that the visual esthetics provided by a mobile augmented reality (AR) service positively influence users' sense of spatial presence, which in turn enhances their flow experience. Lee et al. (2024) further demonstrated that escapism experiences play a crucial role in enhancing flow, indicating that

highly concentrative, immersive, and intrinsically motivating experiences are essential for achieving a flow state.

Flow, in turn, has been shown to significantly influence both educational and entertainment experiences. Previous studies have found that entering a state of flow enhances attitudes toward learning, creativity, innovation, and overall learning outcomes (Agarwal & Karahanna, 2000; Choi et al., 2007; Finneran & Zhang, 2005). Hoffman and Novak argued that flow leads to outcomes such as a positive subjective experience and increased learning. Similarly, Moran (2012) found that high levels of immersion and concentration allow students to tackle complex tasks more effectively, improving problem-solving abilities and promoting more coherent and systematic thinking. Flow has been also found to enhance the hedonic value of online shopping experiences (Senecal et al., 2002) and increase positive emotions and enjoyment (H. Chen, 2006). Ghani and Deshpande (1994) demonstrated a positive connection between flow and media enjoyment, while H. Wang and Lee (2020) found that concentration positively correlates with entertainment in the context of mobile social network games.

Based on these findings, we propose the following hypotheses:

H_5: Flow mediates the relationship between immersion (escapism and esthetic experiences) and absorption (educational and entertainment experiences).

2.3. Satisfaction and intention to visit

Satisfaction is a crucial element in various fields, with H. Zhang et al. (2018) identifying it as a fundamental component of the tourism experience. Satisfaction reflects the overall sentiment that tourists derive from visiting attractions and participating in tourism activities (S. Lee et al., 2020; Li et al., 2023; Wani et al., 2023). VR tour offers a unique opportunity to enhance visitor engagement and provide local information before actual site visits by creating immersive, interactive environments that replicate or even surpass real-world experiences (Bruno et al., 2010; M. J. Kim & Hall, 2019). VR's ability to deliver educational, entertainment, esthetic, and escapism experiences enriches tourists' interactions with destinations, significantly improving the overall quality of tourism experiences (Guttentag, 2010).

Specifically, C. F. Chen and Chen (2010) found that educational experiences significantly influence tourist satisfaction. Similarly, Mehmetoglu and Engen (2011) demonstrated that educational experiences play an important role in enhancing visitors' satisfaction at museums. Additionally, entertainment aspects are known to significantly enhance the overall experience (Crick-Furman & Prentice, 2000). Sotiriadis (2017) observed that a lack of entertainment experiences can reduce satisfaction in wine tourism. Both educational and entertainment experiences are thus crucial in shaping the overall travel experience and visitor satisfaction (Bujisic et al., 2015; T. Jung et al., 2016).

Based on these insights, we propose the following hypotheses:

H_6: Educational experiences have a positive effect on satisfaction.

H_7: Entertainment experiences have a positive effect on satisfaction.

The intention to visit refers to the likelihood of a consumer visiting a destination in person and is a key indicator of behavioral loyalty, particularly in tourism studies (Liang et al., 2021). In tourism research, satisfaction plays a crucial role in shaping tourists' intentions to visit (S. Lee et al., 2020; Wani et al., 2023). Based on these insights, we propose the following hypothesis:

H_8: Satisfaction positively affects the intention to visit the destination in person.

3. Methodology

This study explores the relationship between immersive experiences (escapism and esthetic experiences) and absorptive experiences (educational and entertainment experiences), as well as the impact of absorptive experiences on overall satisfaction and their subsequent impact on the intention to visit a destination. By focusing on three first-order dimensions of flow – enjoyment, concentration, and time distortion – we also examine the mediating effect of flow in the relationships between immersive experiences and absorptive experiences. Our research model is presented in Figure 1.

3.1. Data collection and analysis

To test the proposed model, this study adopts a survey method. Data were collected from June 1 to 14 June 2024, with participants randomly selected from the general population through online recruitment. A total of 350 participants were recruited before the experiment began. No distinction was made between experienced and inexperienced participants during recruitment. During the VR experiment, participants wore either Oculus Quest 2 or Pico 4 VR headsets, with both Korean and Chinese language options provided. Participants accessed the VR homepage by clicking the link https://www.hwasun.go.kr/vr/, where they could choose attractions to explore based on personal preferences. The VR tourism content focuses on Hwasun, a county in Jeollanam-do, South Korea, known for its rich historical and cultural heritage. The VR experiment lasted approximately 20 minutes, during which

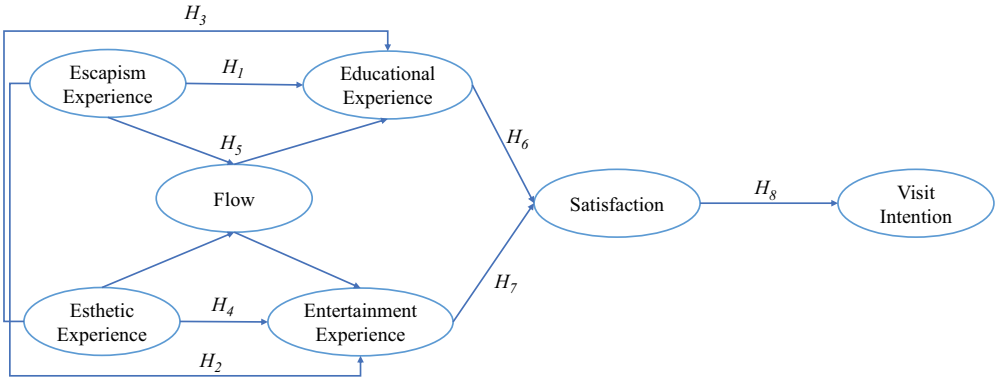


Figure 1. Research Model.

Table 1. Demographic characteristics of respondents.

Item	Characteristics	Samples (350)	Ratio (%)
Sex	Male	178	50.9
	Female	172	49.1
Age	18–25	66	18.9
	26–30	167	47.7
	31–40	87	24.9
	41–50	30	8.6
Educational level	High School Diploma	21	6.0
	Bachelor's Degree	289	82.6
	Master's Degree & Doctorate or higher	40	11.4
Monthly spending power (dollars)	0–165	47	13.4
	165–250–	33	9.4
	250–330	87	24.9
	330–415	102	29.1
	More than 415	81	23.1
Previous VR experience	Yes	275	78.6
	No	75	21.4

participants could freely switch between attractions. Afterward, each participant had a 10-minute break, followed by a 5–10 minute online questionnaire survey. Table 1 details the sociodemographic characteristics of the 350 participants.

3.2. Measurement

The measurements used in this study were adapted from previous studies and modified to fit our context. The measurement items for escapism, esthetic, educational, and entertainment experiences in VR tourism were based on H. Lee et al. (2020) and Song et al. (2015). Items for measuring flow were adapted from Pelet et al. (2017), Leveau and Camus (2023), and Zhu et al. (2020). Satisfaction was measured using items from Song et al. (2015). Intention to visit was assessed based on Jang et al. (2015). All constructs were measured using a five-point Likert scale, ranging 1 (strongly disagree) to 5 (strongly agree).

4. Results

To test the model, this study employed structural equation modeling to examine the statistical significance of the relationships between the proposed variables. Partial least squares (PLS) regression was used for data analysis, as it is particularly suitable for the early stages of theory development (Barclay et al., 1995). We estimate both the measurement and structural model using Smart-PLS 3.

4.1. Measurement model

We assessed the internal validity, as well as the convergent and discriminant validities, of the measurement model. As depicted in Table 2, each factor exhibited Cronbach's alpha values ranging from 0.804 to 0.872, indicating a high level of scale reliability. Composite Reliability (CR) exceeded 0.80, providing robust evidence of high internal consistency (Nunnally & Bernstein, 1994). Additionally, we consider the Average Variance Extracted

Table 2. Internal reliability and convergent validity of the measurements.

Constructs	Items	Factor Loading	Cron-bach's alpha (> 0.7)	CR rho_a (> 0.7)	CR rho_c (> 0.7)	AVE (> 0.5)
Educational experience (EDU)	I learned something new during the VR tour.	0.849	0.852	0.852	0.900	0.693
	The VR tour expanded my knowledge.	0.819				
	It sparked my curiosity to learn new things.	0.857				
	The VR tour provided a great learning experience.	0.804				
Entertain-ment Experience (ENT)	The VR tour was enjoyable.	0.801	0.804	0.804	0.872	0.631
	The VR tour was entertaining.	0.799				
	The VR tour was captivating.	0.838				
	The VR tour was fun.	0.736				
Escapism Experience (ESC)	I felt like I played a different character during the VR tour.	0.829	0.872	0.873	0.913	0.723
	I felt like I was living in a different time or place during the VR tour.	0.852				
	The VR tour allowed me to imagine being someone else.	0.857				
	I completely escaped from reality during the VR tour.	0.862				
Esthetic Experience (EST)	The VR tour I experienced was very impressive.	0.849	0.868	0.868	0.910	0.716
	I was moved by the beautiful details of the VR tour.	0.840				
	The VR tour I experienced stimulated my esthetic senses.	0.864				
	I felt a real sense of harmony in the scenes I saw during the VR tour.	0.832				
Enjoyment (FEJ)	I enjoyed the VR tour.	0.876	0.850	0.851	0.909	0.769
	I still feel good after the VR tour.	0.877				
	I want to experience the VR tour again.	0.878				
Concentra-tion (FCO)	I was deeply focused during the VR tour.	0.877	0.845	0.845	0.906	0.763
	I maintained my concentration throughout the VR tour.	0.880				
	My attention was consistently engaged during the VR tour.	0.864				
Flow (FLW)	-		0.868	0.868	0.895	0.640
Time Distortion (FTD)	During the VR tour, I completely lost tract of time.	0.857	0.833	0.833	0.900	0.749
	During the VR tour, I forgot about my immediate surroundings.	0.864				
	The VR tour allowed me to forget about the realities of the world outside.	0.875				
Intention to Visit (ITV)	After the VR tour, I feel inclined to visit the place offline.	0.898	0.852	0.860	0.910	0.772
	After the VR tour, I become interested in the destinations and want to visit them.	0.877				
	The place I experienced through the VR tour will be my next destination.	0.860				
Satisfaction (SAT)	I am satisfied with the VR tour content I experienced.	0.860	0.835	0.837	0.901	0.752
	I found the details of my VR tour experience to be excellent.	0.865				
	I found the environment details of my VR tour to be excellent.	0.876				

(AVE), with all item loadings exceeding 0.60. AVE assesses the convergent validity of the model, and all values in Table 2 surpass the 0.5 threshold, meeting the AVE standard (Hair et al., 2017).

According to the Fornell-Larcker criterion, the square root of the Average Variance Extracted (AVE) for each latent variable should exceed its correlations with other latent variables. As shown in Table 3, this criterion is met, supporting the discriminant validity of our model.

In addition, we calculated the heterogeneous trait-to-monomorphic property ratio (HTMT) as suggested by Ab Hamid et al. (2017). As shown in Table 4, except for FLW and its first-order dimensions (FEJ at 0.935, FCO at 0.918, and FTD at 0.952), the HTMT values for other variables were below the 0.85 threshold, indicating good discriminant validity.

To further evaluate the validity of these indicators (Petter et al., 2007), we tested the variance inflation factor (VIF) values between FLW and its first-order dimensions (FEJ, FCO, and FTD). Table 5 shows that all VIF values were below 2.300, indicating good validity for flow and autotelic experience, concentration, and immersion.

4.2. Structural model

To verify the hypotheses, the structural model was assessed by evaluating the explanatory power of the constructs and the statistical significance of the proposed paths.

Table 3. Correlation matrix.

Constructs	EDU	ENT	ESC	EST	FAE	FCO	FLW	FIM	ITV	SAT
EDU	**0.832**									
ENT	0.505	**0.794**								
ESC	0.410	0.445	**0.850**							
EST	0.381	0.432	0.439	**0.846**						
FAE	0.488	0.478	0.387	0.391	**0.877**					
FCO	0.471	0.482	0.420	0.416	0.432	**0.874**				
FLW	0.585	0.583	0.526	0.539	0.804	0.785	**0.698**			
FIM	0.444	0.439	0.456	0.487	0.493	0.456	0.811	**0.866**		
ITV	0.405	0.453	0.436	0.418	0.426	0.477	0.534	0.378	**0.878**	
SAT	0.449	0.524	0.423	0.404	0.429	0.453	0.561	0.465	0.400	**0.867**

EDU: Education Experience, ENT: Entertainment Experience, ESC: Escapism Experience, EST: Esthetic Experience, FEJ: Enjoyment, FCO: Concentration, FLW: Flow, FTD: Time Distortion, ITV: Intention to Visit, SAT: Satisfaction.
The square root of the AVE is given along the diagonal.

Table 4. Heterotrait-monotrait ratio.

Constructs	EDU	ENT	ESC	EST	FAE	FCO	FLW	FIM	ITV	SAT
EDU										
ENT	0.609									
ESC	0.475	0.527								
EST	0.443	0.515	0.504							
FAE	0.574	0.579	0.448	0.456						
FCO	0.555	0.584	0.49	0.486	0.509					
FLW	0.68	0.697	0.604	0.621	0.935	0.918				
FIM	0.527	0.533	0.534	0.572	0.585	0.542	0.952			
ITV	0.475	0.547	0.506	0.484	0.501	0.561	0.621	0.45		
SAT	0.531	0.633	0.495	0.474	0.51	0.538	0.658	0.556	0.470	

EDU: Education Experience, ENT: Entertainment Experience, ESC: Escapism Experience, EST: Esthetic Experience, FEJ: Enjoyment, FCO: Concentration, FLW: Flow, FTD: Time Distortion, ITV: Intention to Visit, SAT: Satisfaction.

Table 5. Validity analysis of second-order variables.

	VIF		VIF
FEJ1	2.096	**FLW-FEJ1**	2.211
FEJ2	2.019	**FLW-FEJ2**	2.220
FEJ3	2.105	**FLW-FEJ3**	2.182
FCO1	2.059	**FLW-FCO1**	2.174
FCO2	2.100	**FLW-FCO2**	2.181
FCO3	1.935	**FLW-FCO3**	2.079
FTD1	1.826	**FLW-FTD1**	1.969
FTD2	1.966	**FLW-FTD2**	2.071
FTD3	2.001	**FLW-FTD3**	2.162

FEJ: Enjoyment, FCO: Concentration, FLW: Flow, FTD: Time Distortion, FLW: Flow.

Figure 2 shows the R^2 values for the five endogenous latent variables: EDU (0.359), ENT (0.378), SAT(0.320), ITV(0.160), and FLW(0.395), indicating that our model has a moderate to strong explanatory power for most of the endogenous latent variables. As shown in Table 6 and Figure 2, all hypotheses were supported, except for H_3. Escapism experiences are significantly positively correlated with educational experiences

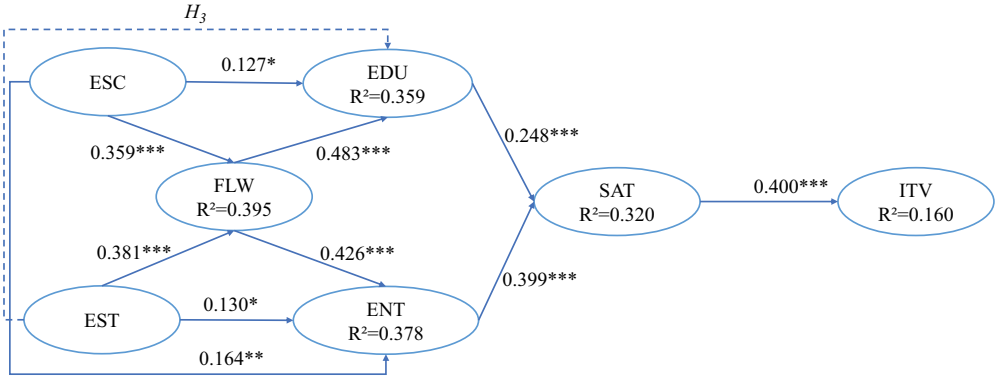


Figure 2. Structural model.

Table 6. Results of PLS path coefficient analysis.

	Original sample (O)	Sample mean (M)	Standard deviation (STDEV)	T statistics
ESC -> EDU (H_1)	0.127*	0.126	0.054	2.361
ESC -> ENT (H_2)	0.164**	0.163	0.051	3.216
EST -> EDU (H_3)	0.065	0.066	0.058	1.127
EST -> ENT (H_4)	0.130*	0.132	0.052	2.494
ESC -> FLW (H_5)	0.359***	0.360	0.042	8.514
EST -> FLW (H_5)	0.381***	0.381	0.045	8.439
FLW -> EDU (H_5)	0.483***	0.483	0.056	8.627
FLW -> ENT (H_5)	0.426***	0.426	0.051	8.406
EDU -> SAT (H_6)	0.248***	0.248	0.048	5.146
ENT -> SAT (H_7)	0.399***	0.399	0.046	8.684
SAT -> ITR (H_8)	0.400***	0.401	0.042	9.464
FLW -> FCO	0.784***	0.784	0.024	33.012
FLW -> FEJ	0.803***	0.803	0.021	38.492
FLW -> FTD	0.812***	0.812	0.020	40.926

EDU: Education Experience, ENT: Entertainment Experience, ESC: Escapism Experience, EST: Esthetic Experience, FEJ: Enjoyment, FCO: Concentration, FLW: Flow, FTD: Time Distortion, ITV: Intention to Visit, SAT: Satisfaction.
*$p < .05$, **$p < .01$, ***$p < .001$.

(coefficient = 0.127*, $p < .05$), and entertainment experiences (coefficient = 0.164**, $p < .01$), supporting H_1 and H_2. On the other hand, while esthetic experiences positively correlate with entertainment experiences (coefficient = 0.130*, $p < .05$), they do not have a significant direct effect on educational experiences. Therefore, H_4 is supported, but H_3 is not.

We examined the mediating effects of flow and found the following: (i) Flow mediates the relationship between escapism experiences and educational experiences, with an indirect effect of 0.173. (ii) Flow mediates the relationship between escapism experiences and entertainment experiences, with a significant indirect effect of 0.152. (iii) Although esthetic experiences do not have a significant direct effect on educational experiences, they have a significant indirect effect via flow, with a path coefficient of 0.184. (iv) Esthetic experiences also have a significant indirect effect on entertainment experiences, mediated by flow, with a path coefficient of 0.162. These results confirm that flow mediates the relationship between immersive and absorptive experiences, supporting H_5.

Additionally, significant positive correlations were observed between educational experiences and satisfaction, as well as between entertainment experiences and satisfaction, supporting H_6 and H_7. We also found a significant positive correlation between satisfaction and visit intention, supporting H_8.

5. Discussion and conclusion

The purpose of this study was to explore the roles of immersion, absorption, and flow in virtual reality tourism, and how they contribute to enhancing customer satisfaction and visit intention. By analyzing the sequential relationship between immersive and absorptive experiences, alongside the mediating role of flow, this study sought to clarify how different types of customer experiences interact within virtual reality tourism context. Furthermore, it aimed to advance beyond the traditional absorption-immersion framework by offering a more nuanced understanding of customer experiences through the interplay of the four experience dimensions and flow.

We found that two immersive experiences played distinct roles in shaping absorptive experiences. As expected, escapism experiences preceded and significantly influenced absorption (i.e. education and entertainment experiences). However, contrary to our expectations, esthetic experiences did not have a significant impact on educational experiences. Since Hypothesis H_3 was not supported, we cannot conclusively state that immersive experiences always precede and influence absorptive experiences. Nevertheless, by incorporating the concept of participation – a key element from the experience economy framework – we can infer from H_1 and H_4 that immersive experiences do precede and influence absorptive experiences, particularly when both experiences align in terms of participation level, whether active or passive. These findings differ considerably from previous studies such as H. Lee et al. (2020) and Park et al. (2010). The contrasting results may be attributed to differences in the variety of experience offered at the target destination (Oh et al., 2007). Alternatively, this discrepancy could arise from the overlapping nature of immersion and absorption, as these two types of experiences often occur simultaneously and cannot be clearly distinguished. Indeed, Pine and Gilmore (1999) suggested each of the four dimensions ultimately combine to form the optimal consumer experience.

Additionally, we found that flow mediates the relationship between immersive experiences and absorptive experiences, regardless of the level of participation. This finding contrasts with the results of Q. Zhang et al. (2021), who reported that flow mediates the relationship between the four experiential dimensions and both word-of-mouth (WOM) and revisit intention. In Q. Zhang et al. (2021), educational and entertainment experiences are considered antecedents of flow, while in our study, they are viewed as consequence of flow. Such inconsistencies in how flow is modeled are commonly observed in the literature. For instance, what one researcher identifies as an antecedent of flow, another may regard as a consequence of flow (Hoffman & Novak, 2009). Indeed, ease of use is considered a consequence of flow in Agarwal and Karahanna (2000), while it is viewed as an antecedent of flow in Hsu and Lu (2003) and Sanchez-Franco (2006). Furthermore, several studies have shown that flow influences attitudes toward websites and brands (Sanchez-Franco, 2006), whereas Luna et al. (2002) modeled a reverse causal sequence, finding that attitudes toward a website can influence flow. These varying perspectives highlight the complexity of flow and underscore the necessity for further research to clarify its role in consumer experiences.

5.1. Implications for fashion marketing theory

This study makes several significant contributions to the field of fashion marketing research. First, it refines Pine and Gilmore's (1999) experience economy theory by elucidating the relationship between immersion and absorption in the context of VR tourism. By incorporating the concept of participation, we demonstrate that immersive experiences significantly influence absorptive experiences in VR when both involve the same level of participation. These findings are particularly relevant for fashion brand marketing, where virtual reality is increasing used in marketing communications. Jung et al. (2021) theorized how consumers derive meanings from VR experiences, suggesting that offering highly immersive experiences can shape and enhance luxury brand communications and consumer behavior. Our study advances the theoretical discourse on VR consumers experiences by focusing on the relationship between absorption and immersion, potentially stimulating further interest among fashion marketing scholars in exploring these constructs within the framework of experience economy.

Second, our study integrates experience economy theory with flow theory to explore the interrelations among the four experiential dimensions. A key finding is that flow mediates the relationship between immersion and absorption, regardless of the level of participation. This underscores the importance of fostering a flow state to enhance both the quality of experiences and brand communications. It offers valuable insights for fashion marketing scholars aiming to understand how flow influences consumer behavior in immersive environments and how to strategically leverage it the marketing communications. Lastly, we measured the flow construct as a second-order variable, comprising enjoyment, concentration, and time distortion. This approach provides a robust framework for assessing and quantifying consumers' flow experiences in virtual reality, offering researchers a valuable tool for investigating VR users' flow states.

5.2. Implications for fashion marketing practice

From a practical standpoint, this study presents several important implications. First, our finding that immersion precedes absorption in virtual reality environments emphasizes the need to deliver high-quality escapism and esthetic experiences when designing VR content. Immersion occurs through passive escapism and esthetic appreciation, where consumers become engrossed in vivid imagery, sounds, and narratives that evoke a sense of a fantasy (Jiang et al., 2023). Virtual reality has expanded consumer experiences into more diverse forms (Lim et al., 2021), allowing for a deeper understanding of consumers' responses to fantasy, fun, and emotional engagement (Rather et al., 2022). This insight helps practitioners optimize VR content design to enhance marketing communication performance, especially in sectors like fashion and luxury marketing. Consumers who experience escapism and esthetic appreciation through virtual reality are more likely to engage with brands, resulting in greater enjoyment in their brand interactions.

Second, our findings indicate that educational and entertainment experiences in virtual reality tourism influence consumer behavioral intentions, such as satisfaction and the intention to visit offline locations. This insight is relevant for other experience-intensive industries, including gaming, beauty, and fashion. Specifically, in fashion marketing, offering virtual reality experiences that enhance consumers' brand knowledge, engagement, and enjoyment can effectively encourage visits to offline brand stores following their virtual experiences (J. Jung et al., 2021). Moreover, our study shows that when flow mediates the relationship, previously insignificant effects of esthetic and educational experiences become statistically significant. This suggests that even in the absence of direct effects, introducing flow as an experiential element can change the dynamics between different types of consumer experiences, underscoring the critical role of flow in experience-based industries.

5.3. Limitations and future research directions

This study has several limitations. First, while it demonstrates the interrelations among immersion, absorption, and flow, these findings are specific to the VR tourism context. Future research should investigate whether similar results can be replicated in other contexts, as well as with different communication tools, such as AR or the metaverse. Second, we measured flow as a second-order construct composed of three dimensions: enjoyment, concentration, and time distortion. The selection of these first-order dimensions was based on existing research and the focus of this study. However, this choice may limit the scope for validating flow's mediating effects. Future research should consider using alternative measures of flow to test the consistency of our findings, irrespective of the specific flow measurement used. Lastly, we did not differentiate between participants with prior VR experience and those without, or account for varying nationalities, which may limit the generalizability of our findings. Additionally, the sample size was relatively small and lacked gender-balance, potentially overlooking gender differences in VR technology usage. Future research should differentiate between VR-experienced and VR-naive participants across different cultures, and conduct group difference analyses to achieve a more comprehensive understanding.

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References

Ab Hamid, M. R., Sami, W., & Sidek, M. H. M. (2017). Discriminant validity assessment: Use of fornell & larcker criterion versus HTMT criterion. *Journal of Physics: Conference Series*, *890*(1), 012163. https://doi.org/10.1088/1742-6596/890/1/012163

Agarwal, R., & Karahanna, E. (2000). Time flies when you're having fun: Cognitive absorption and beliefs about information technology usage. *MIS Quarterly*, *24*(4), 665–694. https://doi.org/10.2307/3250951

Alexander, B., & Rose Thompson, H. (2024). *Customer-centric luxury fashion store experiences: A case study approach*. DRS2024: Boston, USA. https://doi.org/10.21606/drs.2024.237

Barclay, D., Higgins, C., & Thompson, R. (1995). The partial least squares (PLS) approach to causal modeling: Personal computer adoption and use as an illustration. *Technology Studies*, *2*(2), 285–309.

Bruno, F., Bruno, S., De Sensi, G., Luchi, M. L., Mancuso, S., & Muzzupappa, M. (2010). From 3D reconstruction to virtual reality: A complete methodology for digital archaeological exhibition. *Journal of Cultural Heritage*, *11*(1), 42–49. https://doi.org/10.1016/j.culher.2009.02.006

Bujisic, M., Bilgihan, A., & Smith, S. (2015). Relationship between guest experience, personality characteristics, and satisfaction: Moderating effect of extraversion and openness to experience. *Tourism Analysis*, *20*(1), 25–38. https://doi.org/10.3727/108354215X14205687167509

Chang, C. C. (2013). Examining users' intention to continue using social network games: A flow experience perspective. *Telematics and Informatics*, *30*(4), 311–321. https://doi.org/10.1016/j.tele.2012.10.006

Chatterjee, A. (2014). *The aesthetic brain: How we evolved to desire beauty and enjoy art*. Oxford University Press.

Chen, C. F., & Chen, F. S. (2010). Experience quality, perceived value, satisfaction and behavioral intentions for heritage tourists. *Tourism Management*, *31*(1), 29–35. https://doi.org/10.1016/j.tourman.2009.02.008

Chen, H. (2006). Flow on the net-detecting web users' positive affects and their flow states. *Computers in Human Behavior*, *22*(2), 221–233. https://doi.org/10.1016/j.chb.2004.07.001

Choi, D. H., Kim, J., & Kim, S. H. (2007). ERP training with a web-based electronic learning system: The flow theory perspective. *International Journal of Human Computer Studies*, *65*(3), 223–243. https://doi.org/10.1016/j.ijhcs.2006.10.002

Chou, T.-J., & Ting, C.-C. (2003). The role of flow experience in cyber-game addiction. *Cyberpsychology & Behavior*, *6*(6), 663–675. https://doi.org/10.1089/109493103322725469

Crick-Furman, D., & Prentice, R. (2000). Modeling tourists' multiple values. *Annals of Tourism Research*, *27*(1), 69–92. https://doi.org/10.1016/S0160-7383(99)00041-9

Csikszentmihalyi, M. (1975). *Beyond boredom and anxiety*. Jossey-Bass.

Csikszentmihalyi, M. (1997). *Finding flow: The psychology of engagement with everyday life*. Basic Books.

Ding, D. X., Hu, P. J. H., Verma, R., & Wardell, D. G. (2010). The impact of service system design and flow experience on customer satisfaction in online financial services. *Journal of Service Research*, *13*(1), 96–110. https://doi.org/10.1177/1094670509350674

Finneran, C. M., & Zhang, P. (2005). Flow in computer-mediated environments: Promises and challenges. *Communications of the Association for Information Systems*, *15*(1), 4. https://doi.org/10.17705/1cais.01504

Garwood, J. (2020). Escape to learn! An innovative approach to engage students in learning. *The Journal of Nursing Education*, *59*(5), 278–282. https://doi.org/10.3928/01484834-20200422-08

Ghani, J. A., & Deshpande, S. P. (1994). Task characteristics and the experience of optimal flow in human-computer interaction. *The Journal of Psychology*, *128*(4), 381–391. https://doi.org/10.1080/00223980.1994.9712742

Guttentag, D. A. (2010). Virtual reality: Applications and implications for tourism. *Tourism Management*, *31*(5), 637–651. https://doi.org/10.1016/j.tourman.2009.07.003

Hair, J. F., Jr., Matthews, L. M., Matthews, R. L., & Sarstedt, M. (2017). PLS-SEM or CB-SEM: Updated guidelines on which method to use. *International Journal of Multivariate Data Analysis*, *1*(2), 107–123. https://doi.org/10.1504/IJMDA.2017.087624

Han, D. D., Bergs, Y., & Moorhouse, N. (2022). Virtual reality consumer experience escapes: Preparing for the metaverse. *Virtual Reality*, *26*(4), 1443–1458. https://doi.org/10.1007/s10055-022-00641-7

Han, D. D., Boerwinkel, M., Haggis-Burridge, M., & Melissen, F. (2022). Deconstructing immersion in the experience economy framework for immersive dining experiences through mixed reality. *Foods*, *11*(23), 3780. https://doi.org/10.3390/foods11233780

Herrera, V. J. R., Carrillo, E. P. M., Herrera, S. E. V., & Villar, F. R. C. (2020). Influence of experiential marketing on online engagement of the consumer in the fashion industry in the city of Aguascalientes. *Advances in Management and Applied Economics*, *10*(4), 145–171.

Hoffman, D. L., & Novak, T. P. (1996). Marketing in hypermedia computer-mediated environments: Conceptual foundations. *Journal of Marketing*, *60*(3), 50–68. https://doi.org/10.1177/002224299606000304

Hoffman, D. L., & Novak, T. P. (2009). Flow online: Lessons learned and future prospects. *Journal of Interactive Marketing*, *23*(1), 23–34. https://doi.org/10.1016/j.intmar.2008.10.003

Hsu, C.-L., & Lu, H.-P. (2003). Why do people play on-line games? An extended TAM with social influences and flow experience. *Information & Management*, *41*(7), 853–868. https://doi.org/10.1016/j.im.2003.08.014

Huang, H. C., Huang, L. S., Chou, Y. J., & Teng, C. I. (2017). Influence of temperament and character on online gamer loyalty: Perspectives from personality and flow theories. *Computers in Human Behavior*, *70*, 398–406. https://doi.org/10.1016/j.chb.2017.01.009

Huang, S. Y., Kuo, Y. H., & Chen, H. C. (2020). Applying digital escape rooms infused with science teaching in elementary school: Learning performance, learning motivation, and problem-solving ability. *Thinking Skills and Creativity*, *37*, 100681. https://doi.org/10.1016/j.tsc.2020.100681

Irimiás, A., Mitev, A. Z., & Michalkó, G. (2021). Narrative transportation and travel: The mediating role of escapism and immersion. *Tourism Management Perspectives*, *38*, 100793. https://doi.org/10.1016/j.tmp.2021.100793

Jang, S. Y., Chung, J. Y., & Kim, Y. G. (2015). Effects of environmentally friendly perceptions on customers' intentions to visit environmentally friendly restaurants: An extended theory of planned behavior. *Asia Pacific Journal of Tourism Research*, *20*(6), 599–618. https://doi.org/10.1080/10941665.2014.923923

Jiang, Q., Kim, M., Ko, E., & Kim, K. H. (2023). The metaverse experience in luxury brands. *Asia Pacific Journal of Marketing & Logistics*, *35*(10), 250–2520. https://doi.org/10.1108/APJML-09-2022-0752

Jung, J., Yu, J., Seo, Y., & Ko, E. (2021). Consumer experiences of virtual reality: Insights from VR luxury brand fashion shows. *Journal of Business Research*, *130*(SSCI), 517–524. https://doi.org/10.1016/j.jbusres.2019.10.038

Jung, T., Tom Dieck, M. C., Lee, H., & Chung, N. (2016, February 2–5). Effects of virtual reality and augmented reality on visitor experiences in museum. Information and communication technologies in tourism 2016: Proceedings of the international conference in (pp. 621–635). Springer International Publishing, Bilbao, Spain.

Kim, E. J., & Song, N. E. (2022). Effects of AR tour guide application on tourist flow, experiences, and usage intention. *The Journal of the Korea Contents Association*, *22*(2), 487–500.

Kim, K., Chung, T., & Fiore, A. (2022). The influence of interactive social media advertising on consumer experience and fashion-product brand evaluation. *International Textile and Apparel Association Annual Conference Proceedings*, *78*(1). https://doi.org/10.31274/itaa.13692

Kim, M. J., & Hall, C. M. (2019). A hedonic motivation model in virtual reality tourism: Comparing visitors and non-visitors. *International Journal of Information Management*, *46*, 236–249. https://doi.org/10.1016/j.ijinfomgt.2018.11.016

Koufaris, M. (2002). Applying the technology acceptance model and flow theory to on-line consumer behavior. *Information System Research*, *13*(2), 205–223. https://doi.org/10.1287/isre.13.2.205.83

Kozak, M. (2002). Comparative analysis of tourist motivations by nationality and destinations. *Tourism Management*, *23*(3), 221–232. https://doi.org/10.1016/S0261-5177(01)00090-5

Lee, C. M., Dan, H. J., & Cho, M. H. (2024). The impact of freediving activities on flow and self-efficacy by applying experience economy. *Journal of Tourism Enhancement*, *12*(se3), 167–187. https://doi.org/10.35498/kotes.2024.se3.167

Lee, H., Chung, N., & Jung, T. (2015). Examining the cultural differences in acceptance of mobile augmented reality: Comparison of South Korea and Ireland. In I. Tussyadiah & A. Inversini (Eds.), *Information and communication technologies in tourism* (pp. 477–491). Springer.

Lee, H., Jung, T. H., Tom Dieck, M. C., & Chung, N. (2020). Experiencing immersive virtual reality in museums. *Information & Management*, *57*(5), 103229. https://doi.org/10.1016/j.im.2019.103229

Lee, S., Jeong, E., & Qu, K. (2020). Exploring theme park visitors' experience on satisfaction and revisit intention: A utilization of experience economy model. *Journal of Quality Assurance in Hospitality & Tourism*, *21*(4), 474–497. https://doi.org/10.1080/1528008X.2019.1691702

Lee, Y. J., Ha, S., & Johnson, Z. (2019). Antecedents and consequences of flow state in e-commerce. *Journal of Consumer Marketing*, *36*(2), 264–275. https://doi.org/10.1108/JCM-10-2015-1579

Leveau, P. H., & Camus, E. S. (2023). Embodiment, immersion, and enjoyment in virtual reality marketing experiences. *Psychology & Marketing*, *40*(7), 1329–1343. https://doi.org/10.1002/mar.21822

Li, F., Shang, Y., & Su, Q. (2023). The influence of immersion on tourists' satisfaction via perceived attractiveness and happiness. *Tourism Review*, *78*(1), 122–141. https://doi.org/10.1108/TR-02-2022-0078

Liang, A. R. D., Hsiao, T. Y., Chen, D. J., & Lin, J. H. (2021). Agritourism: Experience design, activities, and revisit intention. *Tourism Review*, *76*(5), 1181–1196. https://doi.org/10.1108/TR-10-2019-0438

Lim, H., Childs, M., Cuevas, L., & Lyu, J. (2021). Between you and me: The effects of content ephemerality and the role of social value orientation in luxury brands' social media communication. *Journal of Global Fashion Marketing*, *12*(2), 120–132. https://doi.org/10.1080/20932685.2021.1881579

Lin, A. X., & Jun, J. K. (2022). Examining the relationships among Chinese tourists' theme park experience economy dimensions, memory, flow, and experience satisfaction. *Journal of the Korean Data Analysis Society*, *24*(1), 265–281. https://doi.org/10.37727/jkdas.2022.24.1.265

Luna, D., Peracchio, L. A., & de Juan, M. D. (2002). Cross-cultural and cognitive aspects of web site navigation. *Journal of the Academy of Marketing Sciences*, *30*(4), 397–410. https://doi.org/10.1177/009207002236913

Mehmetoglu, M., & Engen, M. (2011). Pine and Gilmore's concept of experience economy and its dimensions: An empirical examination in tourism. *Journal of Quality Assurance in Hospitality & Tourism*, *12*(4), 237–255. https://doi.org/10.1080/1528008X.2011.541847

Moran, A. (2012). Concentration: Attention and performance. In S.M. Murphy (Ed.), *The oxford handbook of sport and performance psychology* (pp. 117–130). Oxford University Press. https://doi.org/10.1093/oxfordhb/9780199731763.013.0006

Nakamura, J., & Csikszentmihalyi, M. (2009). Flow theory and research. In C.R. Snyder & S.J. Lopez (Eds.), *The oxford handbook of positive psychology* (pp. 195–206). Oxford University Press.

Nasiri, H., Kirillova, K., & Wassler, P. (2024). Beyond beauty: Ugly and borderline tourism aesthetic experiences. *Annals of Tourism Research*, *105*, 103703. https://doi.org/10.1016/j.annals.2023.103703

Nunnally, J. C., & Bernstein, I. H. (1994). *Psychological theory*. McGraw-Hill.

Oh, H., Fiore, A. M., & Jeoung, M. (2007). Measuring experience economy concepts: Tourism applications. *Journal of Travel Research*, *46*(2), 119–132. https://doi.org/10.1177/0047287507304039

Park, M., Oh, H., & Park, J. (2010). Measuring the experience economy of film festival participants. *International Journal of Tourism Sciences*, *10*(2), 35–54. https://doi.org/10.1080/15980634.2010.11434625

Parrish, P. E. (2009). Aesthetic principles for instructional design. *Educational Technology Research & Development*, *57*(4), 511–528. https://doi.org/10.1007/s11423-007-9060-7

Pelet, J. É., Ettis, S., & Cowart, K. (2017). Optimal experience of flow enhanced by telepresence: Evidence from social media use. *Information & Management*, *54*(1), 115–128. https://doi.org/10.1016/j.im.2016.05.001

Petter, S., Straub, D., & Rai, A. (2007). Specifying formative constructs in information systems research. *MIS Quarterly*, *31*(4), 623–656. https://doi.org/10.2307/25148814

Pine, B. J., & Gilmore, H. J. (1999). *The experience economy: Work is theatre & every business a stage*. Harvard Business School Press.

Quadri-Felitti, D., & Fiore, A. M. (2012). Experience economy constructs as a framework for understanding wine tourism. *Journal of Vacation Marketing*, *18*(1), 3–15. https://doi.org/10.1177/1356766711432222

Rather, R. A., Parrey, S. H., Gulzar, R., & Rehman, S. U. (2022). Does gamification effect customer brand engagement and co-creation during pandemic? A moderated-mediation analysis. *Journal of Global Scholars of Marketing Science*, *33*(2), 1–27. https://doi.org/10.1080/21639159.2022.2083000

Sanchez-Franco, M. J. (2006). Exploring the influence of gender on web usage via partial least squares. *Behavior & Information Technology*, *25*(1), 19–36. https://doi.org/10.1080/01449290500124536

Senecal, S., Gharbi, J.-E., & Nantel, J. (2002). The influence of flow on hedonic and utilitarian shopping values. *Advances in Consumer Behavior*, *29*(1), 483–484.

Song, H. J., Lee, C. K., Park, J. A., Hwang, Y. H., & Reisinger, Y. (2015). The influence of tourist experience on perceived value and satisfaction with temple stays: The experience economy theory. *Journal of Travel & Tourism Marketing*, *32*(4), 401–415. https://doi.org/10.1080/10548408.2014.898606

Sotiriadis, M. (2017). Experiential dimensions and their influence on behavioral intentions within the context of nature-based tourism. *Tourism and Hospitality Management*, *23*(1), 5–50. https://doi.org/10.20867/thm.23.1.7

Sweeney, J. C., & Soutar, G. N. (2001). Consumer perceived value: The development of a multiple item scale. *Journal of Retailing*, *77*(2), 203–220. https://doi.org/10.1016/S0022-4359(01)00041-0

Verhagen, T., Feldberg, F., van den Hooff, B., Meents, S., & Merikivi, J. (2012). Understanding users' motivations to engage in virtual worlds: A multipurpose model and empirical testing. *Computers in Human Behavior*, *28*(2), 484–495. https://doi.org/10.1016/j.chb.2011.10.020

Wang, H., & Lee, K. (2020). Getting in the flow together: The role of social presence, perceived enjoyment and concentration on sustainable use intention of mobile social network game. *Sustainability*, *12*(17), 6853. https://doi.org/10.3390/su12176853

Wang, J., Wang, M., & Wu, J. (2015). Empirical study on flow experience in China tourism E-commerce market. *Journal of Industrial Engineering & Management*, *8*(2), 349–364. https://doi.org/10.3926/jiem.1393

Wang, Y., Ko, E., & Wang, H. (2021). Augmented reality (AR) app use in the beauty product industry and consumer purchase intention. *Asia Pacific Journal of Marketing & Logistics*, *34*(1), 110–131. https://doi.org/10.1108/APJML-11-2019-0684

Wani, M. D., Dada, Z. A., & Shah, S. A. (2023). Behavioral intention and satisfaction levels among pilgrim tourists in South Asia. *Tourism Review*, *78*(4), 1147–1163. https://doi.org/10.1108/TR-09-2022-0432

Wiesemüller, P., Mateen, S., Dengel, A., & Voß-Nakkour, S. (2023). Access to escape-an immersive game-based learning experience for accessibility education in virtual reality. International Conference on Immersive Learning (pp. 273–288). Springer Nature Switzerland, Cham.

Zhang, H., Wu, Y., & Buhalis, D. (2018). A model of perceived image, memorable tourism experiences and revisit intention. *Journal of Destination Marketing & Management*, *8*, 326–336. https://doi.org/10.1016/j.jdmm.2017.06.004

Zhang, Q., Liu, X., Li, Z., & Tan, Z. (2021). Multi-experiences in the art performance tourism: Integrating experience economy model with flow theory. *Journal of Travel & Tourism Marketing*, *38*(5), 491–510. https://doi.org/10.1080/10548408.2021.1952148

Zhang, Q., & Xu, H. (2020). Understanding aesthetic experiences in nature-based tourism: The important role of tourists' literary associations. *Journal of Destination Marketing & Management*, *16*, 100429. https://doi.org/10.1016/j.jdmm.2020.100429

Zhu, W., Yan, R., & Ding, Z. (2020). Analyzing impulse purchasing in cross-border electronic commerce. *Industrial Management & Data Systems*, *120*(10), 1959–1974. https://doi.org/10.1108/IMDS-01-2020-0046